

EXAMEFORGE V1 – DETAILED 15-DAY EXECUTION ROADMAP

AI EXAM INTELLIGENCE PLATFORM

DEVELOPER A (YOU) – AI + BACKEND + INFRASTRUCTURE

BECAUSE YOU ALREADY KNOW:

- ✓ PYTHON
- ✓ FASTAPI
- ✓ POSTGRESQL
- ✓ REDIS
- ✓ GENAI (OLLAMA)
- ✓ QDRANT

DEVELOPER B – FRONTEND + UI/UX + CLIENT EXPERIENCE

RESPONSIBILITIES:

- ✓ NEXT.JS
- ✓ TAILWIND
- ✓ ANIMATION
- ✓ FORMS
- ✓ DASHBOARD
- ✓ STATE MANAGEMENT

PHASE 1 – FOUNDATION & OCR PIPELINE (DAYS 1–3)

DEVELOPER A (BACKEND + AI) SETUP:

- ✓ FASTAPI STRUCTURE
- ✓ ENVIRONMENT CONFIG
- ✓ POSTGRESQL (SCHEMAS)
- ✓ REDIS (QUEUES)
- ✓ DOCKER
- ✓ OCR & DOCUMENT APIs

DELIVER:

BACKEND/

|—— AUTH/
|—— USERS/
|—— DOCUMENTS/
|—— OCR/
|—— EMBEDDINGS/
|—— DATABASE/
|—— MIDDLEWARE/
└—— CORE/

TASKS:

- LOGIN API (OTP DIRECT)
- USER PROFILE API
- DOCUMENT UPLOAD API
- OCR PROCESSING QUEUE
- TEXT CLEANING PIPELINE
- DB SCHEMAS (USERS, DOCUMENTS, BOOKS)

DEVELOPER B (FRONTEND)

SETUP:

FRONTEND/

|—— LANDING/
|—— AUTH/
|—— DASHBOARD/
|—— LIBRARY/
|—— COMPONENTS/
|—— ANIMATIONS/
└—— SERVICES/

TASKS:

- LANDING PAGE
- NAVBAR
- LOGIN UI (OTP FLOW)
- PROFILE UI

- DASHBOARD UI
- THEME

DELIVER PAGES:

/

/LOGIN

/PROFILE

/DASHBOARD

PHASE 2 – KNOWLEDGE PROCESSING (DAYS 4–6)

DEVELOPER A

BUILD:

- POST /documents/upload
- GET /documents/:id/status
- OCR PIPELINE (DEWEW, NOISE REMOVAL, CONTRAST)
- TEXT CHUNKING LOGIC
- TOPIC DETECTION (OLLAMA)
- CHAPTER MAPPING

FEATURES:

- OCR QUEUE (REDIS)
- ASYNCHRONOUS PROCESSING
- VECTOR EMBEDDINGS (QDRANT)
- METADATA EXTRACTION

DEVELOPER B

BUILD:

- UPLOAD FLOW
- UPLOADING UI
- PROCESSING STATES
- DOCUMENT LIBRARY
- FILE CARDS
- METADATA VIEW

FRONTEND DELIVER:

UPLOAD

→ PROCESSING

→ COMPLETED

→ LIBRARY VIEW

PHASE 3 – PRODUCT LAYER & SEARCH (DAYS 7–9)

DEVELOPER A

BUILD:

- SEARCH API
- VECTOR SEARCH LOGIC
- NOTES GENERATION ENDPOINT
- FLASHCARD GENERATION ENDPOINT
- WORKSPACE APIs

DATABASE:

- USERS
- DOCUMENTS
- CHAPTERS
- TOPICS
- EMBEDDINGS
- NOTES

DEVELOPER B

BUILD:

- LIBRARY
- WORKSPACE
- SIDEBAR
- DOCUMENT CARDS
- SETTINGS

PAGES:

/DASHBOARD

/LIBRARY

/DOCUMENT/:ID

PHASE 4 – AI CORE (DAYS 10–12)

DEVELOPER A (MAIN OWNER)

BUILD:

- AI CHAT ENGINE
- RAG PIPELINE
- QDRANT INTEGRATION
- OLLAMA PROMPT OPTIMIZATION
- EMBEDDINGS RETRIEVAL

AGENTS:

- SUMMARIZER AGENT
- QUESTION GENERATOR AGENT
- FLASHCARD AGENT
- CHAT AGENT
- PREDICTION AGENT

PIPELINE:

UPLOAD

↓

EXTRACT

↓

CHUNK

↓

EMBED

↓

STORE

↓

RETRIEVE

↓

RESPONSE

DEVELOPER B

BUILD:

- CHAT UI
- STREAMING UI
- NOTES UI
- FLASHCARD UI
- HISTORY
- TYPING INDICATOR

PAGES:

/WORKSPACE/CHAT

/WORKSPACE/NOTES

PHASE 5 – UPI PAYMENTS & DAILY PRACTICE (DAYS 13–14)**USE:****RAZORPAY****DEVELOPER A:**

- ORDERS API
- VERIFICATION LOGIC
- WEBHOOKS HANDLER
- UPI AUTO PAY SETUP
- SUBSCRIPTION PLANS

DEVELOPER B:

- PRICING PAGE
- CHECKOUT UI
- SUCCESS PAGE
- SUBSCRIPTION DASHBOARD

OWNERSHIP RULES**DEVELOPER A OWNS:**

BACKEND/*

DOCKER/*

DATABASE/*

AI/*

DEVELOPER B OWNS:

FRONTEND/*

UI/*

NO EDITING EACH OTHER'S FOLDERS WITHOUT PR.

DAILY PROCESS

MORNING:

- 15 MIN SYNC

WORK:

- SEPARATE BRANCHES

NIGHT:

- MERGE

BRANCHES:

MAIN

DEVELOP

FEATURE/AUTH

FEATURE/LIBRARY

FEATURE/AI

FEATURE/PAYMENT

FIRST MILESTONE (NEXT GOAL)

BY END OF FIRST MILESTONE:

- LANDING PAGE
- OTP LOGIN
- PROFILE
- DOCUMENT UPLOAD
- OCR PROCESSING
- DOCUMENT LIBRARY
- AI NOTES GENERATION
- BASIC CHAT

**THAT'S THE FIRST VERSION THAT STARTS LOOKING LIKE A REAL SaaS PLATFORM
RATHER THAN AN AI UPLOAD DEMO.**